For the film to be anti-reflecting, the reflected rays should interfere destructively. The least optical path difference introduced by the film should be λ. The optical path difference between the waves reflected from the two surfaces of the film is $2\mu d$. For strong reflection, $2\mu d = \lambda$ [As given in equation (7.36). with $n = 1$].

Rewriting, $d = \dfrac{\lambda}{2\mu}$

Substituting, $d = \dfrac{589\times10^{9}}{2\times1.25} = 235.6\times10^{-9}$

$d = 235.6\times10^{-9} = 235.6$ nm

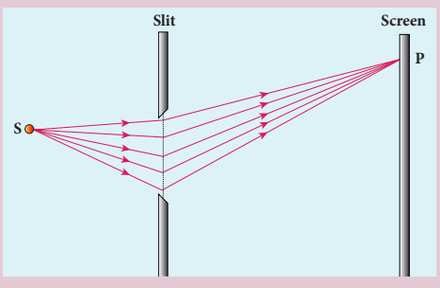
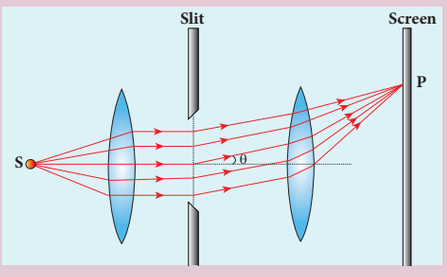
7.4 DIFFRACTION

Diffraction is a characteristic of all waves, including sound waves. ***Diffraction* is bending of waves around sharp edges into the geometrically shadowed region.** This is a violation to the rectilinear propagation of light we have studied in ray optics. But, the diffraction is prominent only when the size of the obstacle is comparable to the wavelength of light. This is the reason why sound waves get diffracted prominently by obstacles like doors, windows, buildings etc. The wavelength of sound wave is large and comparable to the geometry of these obstacles. But the diffraction in light is more pronounced when the obstacle size is of the order of wavelength of light.

7.4.1 Fresnel and Fraunhofer diffractions

Based on the type of wavefront which undergoes diffraction, it could be classified as Fresnel and Fraunhofer diffractions. The differences between Fresnel and Fraunhofer diffractions are shown in Table 7.1.

Table 7.1 Difference between Fresnel and Fraunhofer diffractions

S.No.	Fresnel diffraction	Fraunhofer diffraction
1	Spherical (or) cylindrical wavefront undergoes diffraction	Plane wavefront undergoes diffraction
2	Light wave is from a source at finite distance	Light wave is from a source at infinity
3	Convex lenses need not be used for laboratory conditions	Convex lenses are to be used in laboratory conditions
4	Difficult to observe and analyse	Easy to observe and analyse
5		

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